CHAPTER 7

WEB HOSTING

CHAPTER ESSENCE AT A GLANCE:
Master domain name registration, hosting server selection, DNS mapping, FTP deployment, and distinguishing between shared, VPS, dedicated, and cloud hosting.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Hosting Foundation Fundamentals:** Publishing a website online requires securing a globally unique registered domain name, leasing dedicated server storage space, and establishing reliable high-speed network connectivity.

• **Hosting Tier Architectures:** Hosting solutions range from economical shared servers where multiple tenants divide machine resources, to private virtual slices with guaranteed allocations, up to dedicated servers and scalable distributed cloud architectures.

• **File Deployment Protocol:** Web developers publish local project assets to remote server public directories using authenticated file transfer protocol clients operating across standard control and data channels.

• **Bandwidth and High Availability:** Web hosting capacity depends on monthly data transfer allocations and continuous server uptime guarantees, ensuring uninterrupted accessibility during high-traffic surges.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{Hosted Website} = \text{Domain Name} + \text{Server Space} + \text{FTP Files}$$

Three essential pillars of public website deployment

$$\text{Data Transfer Rate} = \frac{\text{Total Transferred Bits}}{\text{Connection Time}}$$

Hosting bandwidth consumption and throughput relation

$$\text{Shared (Budget)} \rightarrow \text{VPS (Isolated)} \rightarrow \text{Cloud (Elastic)} \rightarrow \text{Dedicated (Peak)}$$

Web hosting tiers ordered by resource isolation and power

$$\text{Availability Percentage} = \frac{\text{Total Uptime Minutes}}{\text{Total Calendar Minutes}} \times 100\%$$

Server Level Agreement (SLA) uptime guarantee formula

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Domain registration reserves name rights globally through ICANN, while web hosting reserves physical server disk storage and bandwidth.

• **Alert:** Shared hosting shares server CPU and RAM among multiple clients, whereas dedicated hosting reserves entire physical machines for a single client.

• **Alert:** FTP authentication credentials must be kept secure; prefer SFTP over port 22 for encrypted file transfer.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is Web Hosting? Explain the essential resources provided by a web host (storage, bandwidth, server uptime). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Web Hosting):** Service providing storage space, server infrastructure, and network connectivity on the internet to make a website accessible 24 divided by 7 globally. **[1 Mark]**

• **Step 2 (Essential Requirements):** 1. Domain Name. 2. Web server space. 3. 24 divided by 7 Internet bandwidth. 4. Web server software (Apache/Nginx). 5. Database support (MySQL). **[1 Mark]**

• **Key Exam Result:** Web hosting: server storage, bandwidth, and 24 divided by 7 global access

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2. Explain the process of domain name registration and the role of ICANN and registrars. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Registration):** Reserving a unique name on the Internet through an accredited registrar (GoDaddy, Namecheap) for an annual fee. **[1 Mark]**

• **Step 2 (TLDs):** Generic TLDs (gTLD): .com (commercial), .org (non-profit), .edu (educational), .gov (government). Country-code TLDs (ccTLD): .in (India), .uk (United Kingdom), .us (United States). **[1 Mark]**

• **Key Exam Result:** Domain registration via ICANN registrars; gTLDs (.com) & ccTLDs (.in)

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Three essential pillars of public website deployment

3. Explain the different types of web hosting: Shared Hosting, VPS Hosting, Dedicated Hosting, and Cloud Hosting. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Shared Hosting):** Multiple websites share resources (CPU, RAM, disk) of a single server. Inexpensive, ideal for small low-traffic sites. **[1 Mark]**

• **Step 2 (VPS Hosting):** Single physical server partitioned into isolated virtual servers using virtualization. Dedicated allocation of RAM/CPU; customizable. **[1 Mark]**

• **Step 3 (Dedicated Hosting):** An entire physical server rented exclusively to one organization. Maximum performance, root access, security; high cost. **[1 Mark]**

• **Step 4 (Cloud Hosting):** Website hosted on a cluster of interconnected virtual servers. Unlimited scalability, zero downtime, pay-as-you-go billing. **[1 Mark]**

• **Key Exam Result:** Shared (low cost) --> VPS (balanced) --> Dedicated (high power) --> Cloud (scalable)

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Three essential pillars of public website deployment

4. Compare Free Web Hosting vs Paid Web Hosting (four points). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Comparison):** Free Hosting: Zero cost, but contains forced advertising banners, slow speeds, zero uptime guarantee, low disk space, and no custom domain. Paid Hosting: Requires subscription, no ads, high bandwidth, 99.9% uptime SLA, 24 divided by 7 support, SSL certificates, custom domain, and automated backups. **[2 Marks]**

• **Key Exam Result:** Free (ad-supported, unreliable) vs Paid (high speed, 99.9% uptime)

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5. Explain the role of FTP (File Transfer Protocol) and FTP client software like FileZilla in web hosting. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (FTP Protocol):** File Transfer Protocol (Port 21) transfers web files from developer's local computer to web hosting server. **[1.5 Marks]**

• **Step 2 (FTP Clients):** Client software (FileZilla, WinSCP) connects via Host, Username, Password, Port. Files dragged and dropped directly into remote /publicHtml/ directory on server. **[1.5 Marks]**

• **Key Exam Result:** FileZilla client uploads files over FTP to /publicHtml/ directory

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Three essential pillars of public website deployment

6. What is cPanel? Explain three essential functions performed using a web hosting control panel. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Role of cPanel):** Web-based graphical control panel interface that simplifies web hosting administration: 1. File Manager (upload, edit files). 2. Database Wizard (create MySQL databases and users). 3. Email Accounts (create domain-based emails info@school.com). 4. Domain management (sub-domains, redirects, DNS zone editor). **[2 Marks]**

• **Key Exam Result:** cPanel: GUI for files, MySQL databases, emails, and DNS records

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7. List the systematic steps to publish a website from local design to live public domain. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Systematic Checklist):** 1. Register custom domain name. 2. Purchase web hosting package (Shared or VPS). 3. Update DNS Name Servers at domain registrar. 4. Upload website files via FTP/cPanel to /publicHtml/. 5. Create MySQL database, import SQL dump, configure connection settings. 6. Install free SSL-TLS certificate (Let's Encrypt). 7. Test links, forms, and mobile responsiveness live. **[3 Marks]**

• **Key Exam Result:** Domain --> Host --> DNS --> FTP Upload --> Database Config --> SSL --> Live Test

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